Authentication and PHY-Security Schemes for Electric Vehicle Dynamic Wireless Charging

Toan-Van Nguyen, *Member, IEEE*, Haijian Sun, *Member, IEEE*, Hongjie Wang, *Senior Member, IEEE*, and Rose Qingyang Hu, *Fellow, IEEE*

***Abstract*—This article proposes authentication and physical layer security schemes to improve secure communications between the electric vehicle (EV) and charging infrastructure in dynamic wireless power transfer (DWPT) systems. In particular, a double-encryption with the signature (DoES) scheme is proposed for session key exchange between EV and charging station which provides data authenticity and integrity. To enable low-latency authentication between EV and power transmitter (PT) in DWPT systems, a sign-encrypt-message (SEM) authentication code scheme is designed leveraging symmetric keys for dynamic charging, which ensures privacy and resistance to tampering attacks. The artificial noise-based physical layer security (AN-based PLS) scheme is also proposed at the physical layer to degrade the wiretapped signal quality of multiple eavesdroppers operating in non-colluding and colluding cases. Closed-form expressions for the secrecy outage probability (SOP) and intercept probability (IP) of the considered system with the non-colluding case are derived to show that the proposed AN-based PLS scheme provides lower SOP and IP than the conventional ones without AN. The distance between eavesdroppers and the PT also affects the system SOP and IP in both non-colluding and colluding cases. Moreover, the EV using the DoES scheme takes 52 ms for obtaining session keys from the charging station while it only spends 8.23 ms with the SEM scheme to authenticate with PT for the charging process.**

***Index Terms*—Authentication, dynamic charging, electric vehicles, privacy, secure wireless networks, and wireless power transfer.**

I. Introduction

RECENTLY, dynamic wireless power transfer (DWPT) has emerged as an innovative technology for a broad range of industrial, service, and household electric vehicles (EVs) since it can efficiently charge the EV while it is in motion. The DWPT technology is a key enabler for EVs to achieve longer driving distances with reduced size of onboard energy storage (battery) [1]. A typical DWPT system is shown in Fig. 1, where the transmitter (primary) pads are placed on the ground side, either on the surface of the roadway or integrated within the roadway with a few inches distance from the roadway surface pavement, while the receiver (secondary) pads are mounted on the bottom of the EV [2]. The electrical power is transferred from the transmitter pads to the receiver pad(s) through magnetic field coupling when the EV passes over the transmitter pads. In dynamic charging systems, the transmitter pads should only switch on during the charging period of authorized EVs to save energy as well as prevent energy theft by unauthorized EVs [3]. To realize such systems, communications among parties such as EV, transmitter pads, the power transmitter (PT), charging station (CS), and road-side unit (RSU) should provide efficient security measurements in a real-time manner which can also protect the private information of the EV drivers, as shown in Fig. 2. However, current wireless communication technologies associated with DWPT systems are still in the early developed stage and expose several security threats to DWPT systems such as man-in-the-middle and eavesdropping attacks. Moreover, through wireless links between the EV and charging infrastructure, the attackers can intercept messages sent by the EV or PT, then fraudulently delay or replays them to somewhere else. Attackers can also launch impersonation attacks by pretending an EV to use dynamic charging without payment or impersonating a CS to hijack sensitive information from the EV and infrastructure to perform network mapping attacks [4]. Lastly, security measurements in DWPT systems should be performed in almost real-time, since the time window

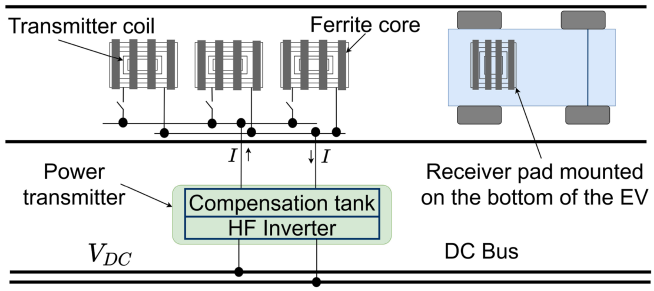


Fig. 1. Typical DWPT system.

Manuscript received 11 February 2023; revised 17 July 2023; accepted 24 August 2023. Date of publication 22 September 2023; date of current version 13 February 2024. The work of Toan-Van Nguyen and Rose Qingyang Hu was supported by the National Science Foundation under the NSF Grants CNS-2007995 and EEC-1941524. The work of Haijian Sun was supported in part by the US National Science Foundation under Grant CNS-2236449, and in part by The University of Georgia E-Mobility Initiative seed grant. The work of Hongjie Wang was supported in part by the US National Science Foundation through the ASPIRE Engineering Research Center and CAREER Award under Grants EEC-1941524 and 204850. The review of this article was coordinated by Dr. Gedare Bloom. *(Corresponding author: Rose Qingyang Hu.)*

Toan-Van Nguyen is with the School of Computer Science and Engineering, International University, Ho Chi Minh 700000, Vietnam, and also with the Vietnam National University, Ho Chi Minh 700000, Vietnam (e-mail: vannguyentoan@gmail.com).

Hongjie Wang and Rose Qingyang Hu are with the Electrical and Computer Engineering Department, Utah State University, Logan, UT 84341 USA (e-mail: hongjie.wang@usu.edu; rose.hu@usu.edu).

Haijian Sun is with the School of Electrical and Computer Engineering, University of Georgia, Athens, GA 30602 USA (e-mail: hsun@uga.edu).

Digital Object Identifier 10.1109/TVT.2023.3318472

0018-9545 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

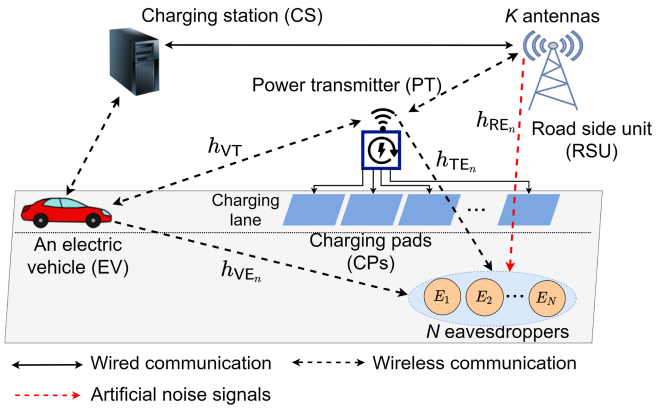


Fig. 2. Considered wireless security DWPT system with multiple eavesdroppers.

for both communication and security schemes is short due to mobility.

A. Related Works

The interaction between EVs and charging entities has exposed several types of attacks such as internal and external attacks, and passive and active attacks, which raise security threats on the confidentiality, authentication, integrity, and availability of EV charging infrastructure [4]. Several countermeasures have been proposed to address such attacks including data authentication, encryption, position verification, and digital certificates. According to the IEEE 802.11p standard, the elliptic curve digital signature algorithm (ECDSA) is used for authentication in-vehicle networks. However, ECDSA spent too much time signing and verifying a signature, which is prone to denial-of-service attacks [5]. In [6], the authors proposed a fast authentication protocol for wireless charging between EVs and charging pads (CPs) relying on symmetric keys and the spatiotemporal location of EVs. In [7], a physical-layer-assisted security scheme for EV dynamic charging system was proposed, where a hierarchical authentication approach with lightweight cryptosystems was developed for authentication among parties such as the bank, CS, RSUs, CPs, and EVs. Previous works mainly focused on the mechanism of power charging management and overlooked data protection against unauthorized access. The introduction of blockchain technology in the DWPT systems allows for addressing these concerns. The distributed consensus process and cryptography employed in blockchain enable it to achieve security and privacy protection [8]. In [9], a private blockchain was designed to support privacy-preserving dynamic charging coordination, authentication, and billing. The proposed strategy introduced the blockchain transactions between the participating entities to reduce overheads and a group signature scheme on top of the blockchain to ensure user anonymity and data unlinkability. Although achieving privacy and security, these blockchain-based approaches involved third-party inclusion, which created the complex exchanged information and delay between the EV and CS. To withstand quantum attacks, a lattice-based signcryption was proposed in [10] for charging security communications which also reduced the latency of cryptographic procedures. However, the charging process was coordinated by multiple servers such as authentication, communication, and processing, which may increase the complexity and scalability of signcryption scheme. In [11], a fast authentication for dynamic EV charging was proposed with fast signing and verification, low communication overhead, and fast hand-off authentication to support EV mobility. The proposed protocol protected the system from replay attacks, MITM attacks, impersonation attacks, EV misreporting, and privacy guarantees. To reduce the complexity of the authentication method, two lightweight mutual authentication schemes between charging plates and the EVs were studied in [12] with a direct authentication with multiple pseudonyms and a hash chain-based authentication. Moreover, different session keys were used in different charging for improving security. A series of legitimate pseudonyms were used instead of real identities to preserve the privacy of the users during the authentication process. In [13], the authors proposed an authentication scheme to improve the security of dynamic charging systems and preserve the privacy of the drivers by combining different cryptosystems such as hashing and Exclusive-OR operations. The proposed scheme could prevent the MITM attack, and attacks against privacy, payment, authentication, and key management.

Considering payment methods for online EVs, a secure and privacy-preserving billing scheme was proposed in [14] with Camenisch-Lysyanskaya signature scheme for guaranteeing the privacy information of users while being against replay attacks. Two mutual authentication mechanisms, namely hash-based authentication and direct authentication, confirmed the communication between EV and the energy segment that can be used for various EV speeds. For enabling a lightweight and secure payment method for dynamic wireless charging, the authors in [15] used low-cost operations including concatenation, XOR, and symmetric cryptography to guarantee security, privacy, anonymity, and resistance to sniff and reply attacks. In [16], a robust and lightweight mutually authenticated key agreement protocol suite with smooth handover was proposed for EV dynamic charging systems. An authentication mechanism for dynamic electric vehicle charging systems based on elliptic curve cryptography, hash functions and chains, concatenation, and exclusive-OR operations was proposed in [17]. Both EV-Auth protocol in [16] and the authentication mechanism in [17] could resist man-in-the-middle (MITM), replay, and insider attacks with user anonymity and untraceability. However, when authentication messages are converted into bits at the physical layer, the eavesdroppers can still overhear and intercept the signals transmitted between the EVs/RSUs and the charging stations, which may expose security threats. Considering blockchain-based dynamic charging, blockchain technology was used for the efficient billing process [18]. Moreover, by leveraging the linear feedback shift registers-based stream cipher, the rapid EV authentication, and authorization for EV dynamic charging were studied. The proposed authentication protocol was shown to resist double-spending issues, replay, and MITM attacks.

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

Relying on a blockchain-based trust model with automated access control of personal rating data, a secure and incentive scheme was proposed in [19] for cooperative DWPT in vehicular energy networks with a hierarchical game-based energy scheduling algorithm. The proposed blockchain-based scheme in [19] approached trust computing, personal data management, and privacy protection in personal rating, which could resist false information injection, advertising fraudulent energy services, and replacing with poor-quality energy. In [20], a lightweight authentication protocol suite was proposed to enable mutual authentication and session key agreement between the EV and the charging system while safeguarding against numerous attacks such as machine learning-based attacks, EV/RSU impersonation, MITM, and insider attacks. In [21], the authors proposed a lightweight authentication and billing scheme for vehicle dynamic charging, where pseudonyms were used to preserve the user's information and supported a single aggregated bill for fair payment. The proposed lightweight authentication protocol is considered to be robust against double-spending attacks, MITM attacks, and impersonation attacks. However, in these previous works, the security at the physical layer has not been studied for wireless charging systems.

Despite the use of encryption methods to ensure the confidentiality of information, eavesdroppers still employ low-level techniques such as analyzing radio frequencies and traffic patterns at the physical layer from the message transmissions of EVs. They can also capture the communication characteristics, such as packet size, duration, period between packets, traffic direction, and channel status information of the wireless channels. By understanding how the protocol works, these attributes can be linked together to create unique fingerprints of the transmissions. Eavesdroppers can then extract a richness of contextual details from these fingerprints, including the importance and locations of the parties involved in the communication, their movement patterns, the applications they use, and the websites they visit. This contextual information can be subsequently exploited to launch advanced attacks such as selective jamming/dropping, targeted node compromise, and DoS attacks. Therefore, the authentication in the application layer is less secure and vulnerable to eavesdropping attacks during the EV charging process.

B. Motivation and Contributions

Designing an authentication scheme for EV wireless dynamic charging is challenging mainly because of the high mobility of EVs. The authentication time should be completed within a short time window (typically a few milliseconds), thus the authentications have to be lightweight and fast, where the digital signature [5] or one-time signature schemes [22] are infeasible for dynamic charging. The CPs typically have limited computational capability and they are not able to run extensive-computation cryptography algorithms [6], [7], [23], [24]. The sign-and-encrypt scheme for authentication in [6], [7] was vulnerable to surreptitious forwarding threats. If a third party is capable of forwarding a pre-existing signed message created by the sender to the receiver then the receiver assumes it is safe to trust this message [25]. The encrypt-and-sign is not better than the sign-and-encrypt scheme since it is easy for an eavesdropper (Eve) to replace the sender's signature with his own, to claim authorship of the encrypted message [25], [26]. Such encryption schemes may improve the security and privacy of wireless transmission only at the upper layers of the protocol stack.

As the number of electric vehicles (EVs), roadside units (RSUs), and public transportation systems (PTs) continue to grow, especially in large-scale networks with dynamic topologies, the process of generating public keys becomes increasingly time-consuming. Additionally, the distribution and management of secret keys become extremely difficult. Moreover, the exchanged information will be fully intercepted if the secret keys are exposed. Different from the cryptographic methods at upper layers, physical layer security (PLS) leverages advantages of the intrinsic characteristics of wireless links, such as interference, noise, and fading, to degrade the received signal qualities of the eavesdroppers, and realize secure keyless communication through signal processing approaches. Thus, PLS can be viewed as an effective approach for security enhancement in DWPT. However, the research on PHY-security and authentication in DWPT systems is yet available in the existing literature. In this article, we propose an artificial noise-based PLS (AN-based PLS) scheme to prevent eavesdropping attacks, which mitigates the probability of wiretapping legitimate wireless channels and can protect the EV's communication during the charging process. The main contributions of the article can be summarized as follows:

- We propose a double encryption and signature (DoES) scheme to protect session keys exchanging among the EV, CS, RSU, and PT from various passive attacks. We show that the DoES scheme provides better security for communications than the sign-and-encrypt in [6].
- To enable low-latency authentication between the EV and PT, we design a sign-encrypt-message authentication code (MAC), namely SEM, scheme for improving the entity authentication, data origin, and integrity during the EV charging process.
- To add an additional layer of security in DWPT systems, we propose an AN-based PLS scheme for enhancing secure performance by using artificial signals radiated from a multi-antenna RSU, which prevents active attacks before EVs reach charging lanes.
- Numerical results show that the DoES scheme takes 52 ms for session key exchange while the SEM scheme only spends 8.23 ms for low-latency authentication. Moreover, the proposed AN-based PLS with non-colluding eavesdroppers provides the lowest SOP and IP among considered schemes, which is a promising scheme for DWPT systems.

Mathematical Notations and Functions: $\Pr[.]$ symbolizes probability, $f_X(.)$ and $F_X(.)$ denote the probability density function (PDF) and cumulative distribution function (CDF) of a random variable X, respectively, and $\mathrm{Ei}(.)$ represents the exponential integral function.

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

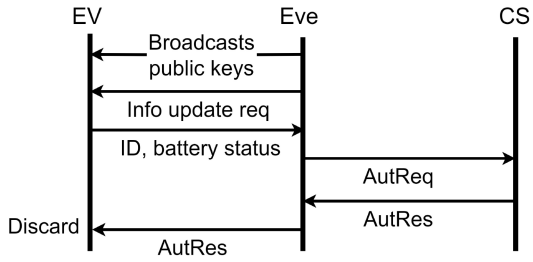


Fig. 3. Attack model on the sign-and-encrypt method.

II. System Model

A. System Description

We consider a dynamic charging scenario, as shown in Fig. 2, where a CS and an RSU are deployed to support the dynamic wireless charging process for an EV via multiple CPs installed underneath the road. An inductive-wireless power transfer is employed for DWPT systems [27], where a PT receives the alternating current (AC) source from the power grid and provides inductive AC links in parallel plate structures. The received AC power at the EV side is then converted to the direct current for charging its battery.

The coverage of CS can be extended to several miles, thus exchanged messages between the CS and RSU are through wired/Internet communication while the communication between EV and CS (or PT) is through wireless cellular networks such as 5 G. Also, the communication between EV and CP is achieved via a dedicated short-range communication device installed at the bottom of the EV. We assume that each EV has a bank account to pay for dynamic charging and anonymously authenticating the RSU to CS [7], [28]. Multiple eavesdroppers can be seen as a group of vehicles moving at the same speed as the EV and they operate on colluding and on-colluding modes to hijack or wiretap the legitimate communication between the EV and CS (or PT). In this work, we design authentication schemes in conjunction with an AN-based PLS scheme to provide high-security performance considering the physical and upper layers in DWPT systems.

B. Attack Model

The asymmetric encryption for exchanging session keys in [6], [7] was shown to withstand impersonate and replay attacks by using the sign-and-encrypt method. However, we will show that such a method does not secure during exchanging session keys process in some specific scenarios. In this attack model, we consider the worst case of non-colluding mode, where the strongest Eve wiretaps the legitimate channels used to exchange messages between the EV and CS. Particularly, when the EV and CS are establishing a secure communication to exchange session keys for dynamic charging, the strongest Eve tries to eavesdrop on these exchanged keys to gain benefits or impersonate one of the parties to confuse the communications and fail the authentication process. Fig. 3 shows the attack procedure which can be presented by the following steps:

i) The Eve broadcasts its public keys to its surrounding EVs with a strong signal power to make these EVs believe the public key belongs to the nearest CS.
ii) To collect information about EVs, Eve now plays the malicious CS and keeps sending the information update request messages to any surrounding EVs.
iii) If an EV within this range needs to update its battery status to the CS for the coming charging, the EV will include its identity and battery status in an information update response, as shown in Fig. 3, which is signed by EV's private key and encrypted by Eve's public key before sending to the malicious CS. In this way, Eve can decrypt the received messages to obtain the identity while still keeping the signed message from the EV.
iv) The Eve uses the CS's public key to re-encrypt this signed message consisting of the identity and battery status of the EV, then sends it to the real CS as the authentication request. Upon arriving at the CS, the message will be decrypted by CS's private key and verified signature by EV's public key to ensure the message from the EV. Since the received message has been signed by the EV, the CS believes the EV sent him directly and cannot detect Eve's fraud, eventually, the signature verification at the CS is successful.
v) The CS then generates the authentication response encompassing session keys and sends it to the EV. The CS also distributes these session keys to the power transmitter and releases these keys into its memory.

Implications: When the real EV needs to charge its battery by sending the authentication request to the CS, the authentication request may be refused because the CS has already generated session keys for this EV associated with the previous identity. Moreover, if Eve constructs and keeps sending a large number of authentication requests with obtained identities, the CS may be busy verifying the signatures and generating the session keys for the authentication responses, which causes the service busy when other EVs need to authenticate with that CS.

C. The Proposed Double Encryption With Signature Scheme

We propose a double encryption and signature scheme for the EV-CS authentication process, as shown in Fig. 4. On the sender side, the sender's message in plain text is first encrypted by the receiver's public inner key to obtain the ciphertext. Subsequently, the cipher text is passed through the hash algorithm block and then encrypted with the sender's private key using cryptographic algorithms to obtain the digital signature. To improve the security of message transmission, an outer encryption layer is employed, where the ciphertext and the signature are encrypted by the receiver's public outer key before sending over wireless channels.

On the receiver side, after detaching the main message and signature, the outer decryption for these ciphertexts is performed with the receiver's private outer key. In the signature verification process, the same cryptographic algorithm is used as the sender side, where the ciphertext of the main message is decrypted by using the sender's private key. Meanwhile, the ciphertext of the

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

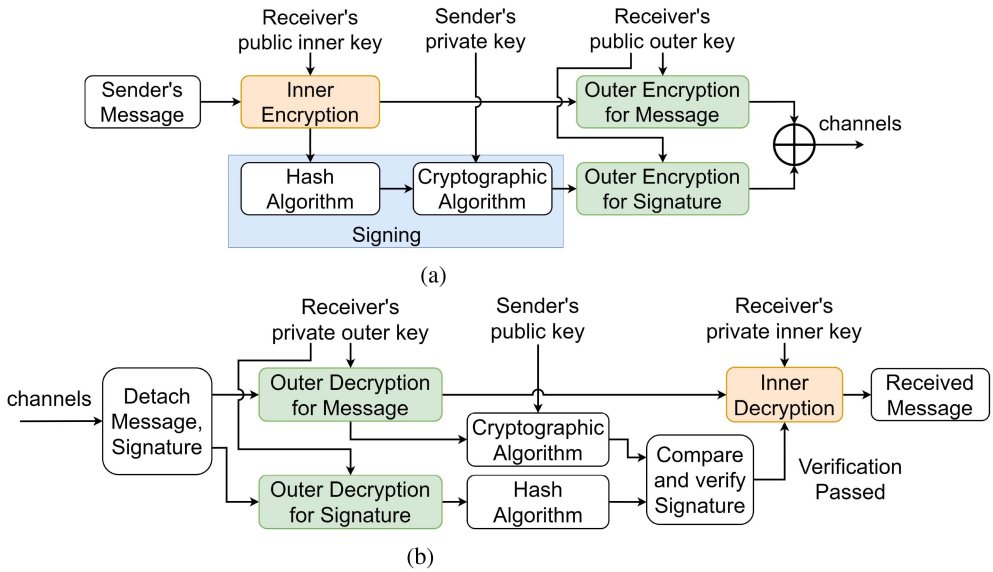


Fig. 4. Proposed double encrypt and sign (DoES) scheme for EV-CS authentication. (a) Sender Side. (b) Receiver Side.

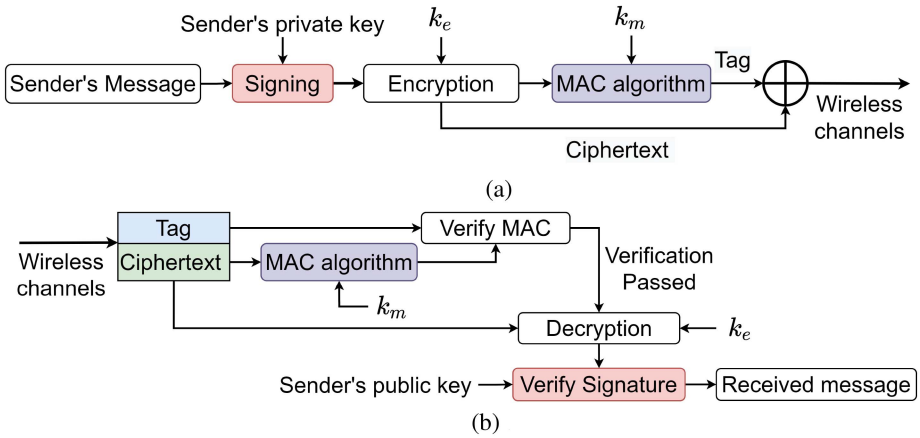


Fig. 5. Proposed *sign-encrypt-MAC* (SEM) scheme for EV-PT authentication. (a) Sender Side. (b) Receiver Side.

signature is passed through the hash algorithm block, where the obtained ciphertext is compared with that of the cryptographic algorithm block. If the signature verification succeeds, the result enables the inner decryption to decrypt the ciphertext of the message to recover the original plaintext. The new features of the proposed DoES are to encrypt the ciphertext and the signature of the ciphertext separately, which reduces the execution time compared to those encryption schemes in [25]. Moreover, making use of two different public keys for double encryption processes can increase the degree of security level for wireless charging. Compared to the sign-and-encrypt scheme in [6], [7], the proposed DoES offers an extra outer encryption layer to prevent the impersonate attack which will be analyzed in the next subsection.

D. The Proposed Sign-Encrypt-MAC Scheme

We propose a sign-encrypt-MAC scheme for authentication between EV and PT before the dynamic charging, as shown in Fig. 5. On the sender side, the message is first signed by the sender's private key, symmetrically encrypted by k_e, and then applied MAC algorithm to create a tag for the ciphertext. The tag and ciphertext are combined before sending over wireless channels. On the receiver side, the received message is detached into the ciphertext and transmitter tag. The MAC algorithm is used to calculate the receiver tag of ciphertext by using key k_m. The transmitter tag is checked by comparing it with the receiver tag in the verification block. If they are identical, the MAC verification will enable the decryption block to decrypt the ciphertext. Otherwise, the ciphertext has been tampered with during the transmission, leading to the failure of MAC verification. The receiver also verifies the signature by the sender's public key before recovering the original plain text.

The encrypt-and-MAC scheme was studied in [6], [7] for the authentication between EV and charging pads. However, such a scheme cannot provide entity authentication since the MAC algorithm just guarantees the message authentication to confirm the origin and integrity of data. In the worst case, it is easy for

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

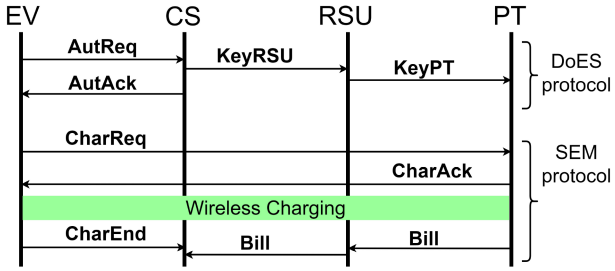


Fig. 6. Charging flow messages in DWPT systems.

anyone obtaining the session keys and identity information to successfully perform authentication with the PT. To tackle this problem, we propose a signing process before encrypting and MAC to achieve entity authentication, which ensures the identity of an EV interacting with the PT during that communication session. *How can the proposed DoES resist the attack discussed in Section II-B?*

By employing the proposed DoES method, at step (ii), the information update message is first encrypted by Eve's public key, then signed by EV's private key, and finally outer encrypted by Eve's public key. Note that the public key for the first encryption is different from the outer encryption to enhance the degree of security level. At step (iii), the Eve performs outer decryption for the received message and has to remove the signature of EV before inner decrypting the message. After obtaining the identity and battery status of the EV, Eve constructs the authentication request, which is first encrypted by CS's public key, signed by Eve's private key, and outer encrypted by another CS's public key. Once receiving the message, the CS decrypts it and fails to verify the signature since the signature did not come from the EV. Thus, the CS ignores the received message even if the identity of the EV is valid.

E. EV-CS Authentication

The message exchanged among parties in DWPT systems is shown in Fig. 6, where the DoES and SEM schemes are used for mutual authentication between the EV and CS, and between the EV and PT, respectively. Mutual authentication is established between EV and CS with integrity protection to ensure the high-security level of exchanging session keys process the first time when EV is approaching the nearest CS. When draining out the battery, the EV requests the nearest CS to get charged by obtaining the secret session keys. The EV-CS authentication can be summarized by the following steps:

- EV estimates the demand charging energy by using its on-board unit as $E_e = (\text{SoC}_d - \text{SoC}_c)E_{\max}$, where $\text{SoC}_d, \text{SoC}_c$, and $E_{\max}$ are the desired state of charge, the current state of charge, and the maximum battery capacity of EV, respectively. The unit of SOC is a percentage point, e.g., $\text{SoC} = 0\%$ represents a depleted battery status and $\text{SoC} = 100\%$ indicates full of energy.
- EV generates and sends an authentication request message $\mathbf{AutReq} : (ID_e, E_e, t_e)$ to the CS, where ID_e and t_e are the identity and the timestamp of EV, respectively. By using the DoES scheme, the encrypted message at the EV can be expressed as

$$\mathbf{AutReq}_{\mathsf{D}} = \mathsf{AEn}\Big(\mathsf{AEn}\Big(\mathbf{AutReq}, k^{\text{CS}}_{\text{pub,in}}\Big), k^{\text{CS}}_{\text{pub,out}}\Big) + \mathsf{AEn}\Big(\mathsf{Sign}\Big(\mathsf{AEn}\Big(\mathbf{AutReq}, k^{\text{CS}}_{\text{pub,in}}\Big), k^{\text{EV}}_{\text{pri}}\Big), k^{\text{CS}}_{\text{pub,out}}\Big), \tag{1}$$

where $\mathsf{AEn}(.)$ and $\mathsf{Sign}(.)$ denote the asymmetric encryption function and sign function, respectively.

- On receiving the message from EV, the CS follows the decryption process at the receiver side of the DoES scheme to check the integrity $\mathbf{AutReq}$. The CS also verifies the timestamp to ensure a small delay during this process. If the timestamp is out of the valid range, the authentication will be aborted. The CS then computes $h(ID_e)$ and compares it with the stored one in the local list, where $h(.)$ denotes the one-way hash function. If the comparison of identity is passed, the authentication will be continued.
- The CS estimates the amount of payment M_e for a given amount of requested energy E_e. It also generates a set of random numbers $\mathcal{K}$ and selects randomly $\{k_e, k_m\} \in \mathcal{K}$ for the symmetric encryption and MAC, respectively. The CS then sends an acknowledgment message containing the identities of CS and EV, session keys, the estimated amount of payment, and the CS's timestamp via the message $\mathbf{AutAck} : (ID_c, ID_e, k_e, k_m, M_e, t_c)$ to the EV, which is encrypted by using DoES scheme as

$$\mathbf{AutAck}_{\mathsf{D}} = \mathsf{AEn}\Big(\mathsf{AEn}\Big(\mathbf{AutAck}, k^{\text{EV}}_{\text{pub,in}}\Big), k^{\text{EV}}_{\text{pub,out}}\Big) + \mathsf{AEn}\Big(\mathsf{Sign}\Big(\mathsf{AEn}\Big(\mathbf{AutAck}, k^{\text{EV}}_{\text{pub,in}}\Big), k^{\text{CS}}_{\text{pri}}\Big), k^{\text{EV}}_{\text{pub,out}}\Big). \tag{2}$$

- After receiving the message, EV decrypts the message by using its private outer key, verifies the signature by the CS's public key, and decrypts by using its private inner key. EV then checks the time t_c within a specific period and the identities of the CS and itself. After passing all these check-up processes, the EV stores these keys locally for authentication with the PT when approaching charging lanes.

On the CS side, based on the requested charging energy E_e, the CS calculates the number of CPs to be activated, denoted by N_{act}, with $N_{act} \leq N_{CP}$, which is expressed as

$$N_{act} = \frac{E_n v_n}{r_C l_C}, \tag{3}$$

where v_n, l_C, and r_C denote the driving speed on the charging lane, the length of each CP, and the rated power (in watts) of CP, respectively. The CS then sends the key distribution message $\mathbf{KeyRSU} : (ID_c, k_e, k_m, N_{act}, t_c)$ containing the session keys for dynamic charging and the number of active CPs to the PT via RSU. The encrypted message at the CS can be expressed as

$$\mathbf{KeyRSU}_{\mathsf{D}} = \mathsf{AEn}\Big(\mathsf{AEn}\Big(\mathbf{KeyRSU}, k^{\text{RSU}}_{\text{pub,in}}\Big), k^{\text{RSU}}_{\text{pub,out}}\Big) + \mathsf{AEn}\Big(\mathsf{Sign}\Big(\mathsf{AEn}\Big(\mathbf{KeyRSU}, k^{\text{RSU}}_{\text{pub,in}}\Big), k^{\text{CS}}_{\text{pri}}\Big), k^{\text{RSU}}_{\text{pub,out}}\Big). \tag{4}$$

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

The timestamp in each message can guarantee the delay requirement for transmitting messages. When receiving the message, the RSU performs the outer decryption and verifies the signature by CS's public key to guarantee that the message comes from the CS, then decrypts the ciphertext by using its private inner key. The RSU also checks timestamps within a specific time period and the temporary identity of CS. RSU continues to disseminate keys via message $\mathbf{KeyPT}: (ID_c, ID_r, k_e, k_m, t_r)$ to the PT under its control, where ID_r and t_r denote the identity and current time of RSU, respectively. $\mathbf{KeyPT}$ message is encrypted by using the DoES scheme as

$$\mathbf{KeyPT}_{\mathsf{D}} = \mathsf{AEn}\Big(\mathsf{AEn}\Big(\mathbf{KeyPT}, k^{\mathrm{PT}}_{\mathrm{pub,in}}\Big), k^{\mathrm{PT}}_{\mathrm{pub,out}}\Big) + \mathsf{AEn}\Big(\mathsf{Sign}\Big(\mathsf{AEn}\Big(\mathbf{KeyPT}, k^{\mathrm{PT}}_{\mathrm{pub,in}}\Big), k^{\mathrm{RSU}}_{\mathrm{pri}}\Big), k^{\mathrm{PT}}_{\mathrm{pub,out}}\Big). \tag{5}$$

Once receiving message $\mathbf{KeyPT}$, the PT performs the recovery process as shown in Fig. 4(b), and stores these keys in its memory for mutual online authentication with the moving EV. At the end of the EV-CS authentication phase, both the EV and the PT obtain the same secret keys to use in a session of charging. After signature verification for the $\mathbf{AutReq}$ message, the CS will send $\mathbf{KeyRSU}$ to RSU, which in turn, sends the $\mathbf{KeyPT}$ to the PT. If the signature verification of $\mathbf{AutReq}$ message is failed, the CS does not need to send the $\mathbf{KeyRSU}$ to the RSU, which can save time and resources.

F. EV Dynamic Charging

When approaching charging lanes, the EV keeps broadcasting the charging request message $\mathbf{CharReq}: (ID_e, l_e, v_e, t_e)$ to the PT, where l_e presents the location of EV at time t_e and v_e is the speed of EV. By using the SEM scheme, the output message at the EV can be expressed as

$$\mathbf{CharReq}_{\mathsf{S}} = \mathsf{En}\Big(\mathsf{Sign}\Big(\mathbf{CharReq}, k^{\mathrm{EV}}_{\mathrm{pri}}\Big), k_e, \mathrm{iv}\Big) + \mathsf{MAC}\Big(\mathsf{En}\Big(\mathsf{Sign}\Big(\mathbf{CharReq}, k^{\mathrm{EV}}_{\mathrm{pri}}\Big), k_e, \mathrm{iv}\Big), k_m\Big), \tag{6}$$

where $k^{\mathrm{EV}}_{\mathrm{pri}}$ is the private key of EV, and iv denotes the initialization vector to guarantee that different encrypted values are achieved when encrypting the same value multiple times, even with the same session secret key. Using iv can add an extra layer of security to the symmetric encryption protocol. Based on the EV's speed and the radio range of devices, the broadcast frequency can be adjusted to guarantee a high probability of successful authentication at the EV while saving energy. For example, if the range of the wireless device is 0.5 m, the vertical distance from the ground CPs to the pads installed at the bottom of the EV is 0.3 m, an EV moving with 100 km/h may broadcast beacons every 14.8 ms [6]. When the PT receives the charging request message in its communication range, the received message will be processed with integrity verification and authenticity, as shown in Fig. 5(b). If the verification and decryption are successful, the PT checks the valid time and location of EV such that $|t_p - t_e| \leq \epsilon_t$, where ϵ_t is the accepted time mismatch. Based on the requested charging, the PT acknowledges by sending the message $\mathbf{CharAck}: (ID_p, l_{cp}, s_{adj}, t_p)$ to the EV and the dynamic wireless charging process will be started, where ID_p, l_{cp}, s_{adj}, and t_p are the identity of PT, the location of charging pads, the speed adjustment information for EV before charging, and the time stamp of the PT, respectively. The $\mathbf{CharAck}$ message at the PT can be encrypted by using the SEM scheme as

$$\mathbf{CharAck}_{\mathsf{S}} = \mathsf{En}\Big(\mathsf{Sign}\Big(\mathbf{CharAck}, k^{\mathrm{PT}}_{\mathrm{pri}}\Big), k_e, \mathrm{iv}\Big) + \mathsf{MAC}\Big(\mathsf{En}\Big(\mathsf{Sign}\Big(\mathbf{CharAck}, k^{\mathrm{PT}}_{\mathrm{pri}}\Big), k_e, \mathrm{iv}\Big), k_m\Big), \tag{7}$$

where $k^{\mathrm{PT}}_{\mathrm{pri}}$ is the private key of the PT. Once the charging is finished, EV notifies CS that the charging process is completed by sending a message $\mathbf{CharEnd}: (ID_e, t_e)$.

$$\mathbf{CharEnd}_{\mathsf{S}} = \mathsf{En}\Big(\mathsf{Sign}\Big(\mathbf{CharEnd}, k^{\mathrm{EV}}_{\mathrm{pri}}\Big), k_e, \mathrm{iv}\Big) + \mathsf{MAC}\Big(\mathsf{En}\Big(\mathsf{Sign}\Big(\mathbf{CharEnd}, k^{\mathrm{EV}}_{\mathrm{pri}}\Big), k_e, \mathrm{iv}\Big), k_m\Big), \tag{8}$$

At the end of the charging process, the actual amount of energy will be sent to the CS and the CS will charge the fees of the EV corresponding to the EV's identity in the EV-CS authentication phase. The mechanism of billing is presented in the following subsection.

G. Billing Phase

The PT does not directly perform the EV billing service but it informs the amount of energy charged for the EV to the CS that the EV has a contract with. At the end of the charging process, the PT calculates the total supplied energy over N_{act} CPs $E_e = \sum_n^{N_{act}} E_n$, where E_n is the amount of energy at each CP for dynamically charging EV. Then PT sends a billing completed message to the CS, i.e., $\mathbf{Bill}: (ID_e, E_e, t_0, t_1)$ for pricing, where t_0 and t_1 are the begin and end time that the first CP charged the EV. The RSU decrypts the billing message and continues encrypting the received message by using SEM scheme. The billing message is encrypted by SEM as follows:

$$\mathbf{Bill}_{\mathsf{S}} = \mathsf{En}\Big(\mathsf{Sign}\Big(\mathbf{Bill}, k^{\mathrm{PT}}_{\mathrm{pri}}\Big), k_e, \mathrm{iv}\Big) + \mathsf{MAC}\Big(\mathsf{En}\Big(\mathsf{Sign}\Big(\mathbf{Bill}, k^{\mathrm{PT}}_{\mathrm{pri}}\Big), k_e, \mathrm{iv}\Big), k_m\Big). \tag{9}$$

During the charging process, the EV keeps broadcasting its message to the power transmitter for the authentication process, which exposes the high risk of security threats at the physical layer such as spoofing and eavesdropping attacks. Security at the physical layer is an effective approach for such a problem, which will be presented in the following section.

III. Artificial Noise-Based Physical Layer Security Scheme

The DWPT system relying on an encryption/decryption approach is not sufficient to ensure information security and becomes more complex when handling a massive number of vehicles and increasingly dynamic wireless charging services. The complexity of generating private-public keys and the difficulty of distribution and management of secret keys also hinder the effectiveness of cryptographic methods, especially in high-dynamic environments. For example, the encryption/decryption

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

in the application layer cannot prevent the EVs from eavesdropping issues in the physical layer of the protocol stack, due to the broadcast characteristic of wireless channels as attackers are traveling at the same speed as the EV to capture their transmitted signals. PLS is an effective solution for such a problem that yields low computational complexity and provides flexible security levels through adaptive transmission design and signal processing techniques. The PLS can be considered as an additional layer of security to DWPT systems, which requires relatively light signal processing operations with minor additional overheads. Therefore, we propose a PLS scheme to enhance the data transmission between the EV and PT before they perform charging at the charging lanes. Once the exchanged session keys phase is finished, CS enables the RSU to generate artificial noise to degrade the received signal qualities at the eavesdropper.

A. Wireless Model

As shown in Fig. 2, we denote by h_{XY} the channel coefficient between two nodes X and Y, where $\mathrm{X} \in \{\mathrm{V}, \mathrm{T}, \mathrm{R}\}$ and $\mathrm{Y} \in \{\mathrm{T}, \mathrm{V}, \mathrm{E_n}\}$, wherein V, T, R, and E_n denote the EV, power transmitter, RSU, and the n-th Eve, respectively. We assume that all wireless channels are Rayleigh fading and the channel between two terminals is reciprocal [29]. The location of eavesdroppers can be estimated by using the EarFisher method in [30], where the EarFisher stimulates wireless eavesdroppers using bait network traffic and then captures eavesdroppers' responses by sensing and analyzing their memory. When the CS sends session keys to the EV, it then also sends a message informing the RSU to activate jamming signals, where the RSU with K antennas generates artificial noises to interfere with the eavesdropper. The received signals at the PT and n-th Eve can be expressed, respectively, as

$$y_{\mathrm{T}} = \sqrt{P_{\mathrm{V}}} h_{\mathrm{VT}} s_{\mathrm{V}} + n_{\mathrm{T}}, \tag{10}$$

$$y_{\mathrm{E}_n} = \sqrt{P_{\mathrm{V}}} h_{\mathrm{VE}_n} s_{\mathrm{V}} + \sum_{k=1}^{K} \sqrt{P_{\mathrm{R}}} h_{\mathrm{RE}_n} s_{\mathrm{R}} + n_{\mathrm{E}_n}, \tag{11}$$

where s_{V} is the transmit signal from EV representing the charging request message **ReqChar** and s_{R} is the jamming signal from RSU. The achievable rate of the $\mathrm{EV} - \mathrm{PT}$ link can be expressed as

$$R_{\mathrm{VT}} = \log_2(1 + \Psi_{\mathrm{V}} \gamma_{\mathrm{VT}}), \tag{12}$$

where $\Psi_{\mathrm{V}} = P_{\mathrm{V}}/\sigma^2$, and σ^2 denotes noise variance. Since multiple eavesdroppers are deployed, they can work independently (non-colluding), or collude. In the non-colluding scenario, the worst case is that the strongest Eve wiretaps the legitimate channels, and the achievable rate of the $\mathrm{EV} - \mathrm{Eve}$ link is expressed as [31]

$$R_{\mathrm{VE}}^{nc} = \log_2 \left(1 + \max_{n=1,\ldots,N} \left\{ \frac{\Psi_{\mathrm{V}} \gamma_{\mathrm{VE}_n}}{\sum_{k=1}^{K} \Psi_{\mathrm{R}} \gamma_{\mathrm{RE}_n} + 1} \right\} \right), \tag{13}$$

where $\Psi_{\mathrm{R}} \triangleq P_{\mathrm{R}}/\sigma^2$ and the superscript nc denotes the non-colluding.

If multiple eavesdroppers collude, the achievable rate of the $\mathrm{EV} - \mathrm{Eve}$ link is expressed as

$$R_{\mathrm{VE}}^{co} = \log_2 \left(1 + \sum_{n}^{N} \left\{ \frac{\Psi_{\mathrm{V}} \gamma_{\mathrm{VE}_n}}{\sum_{k=1}^{K} \Psi_{\mathrm{R}} \gamma_{\mathrm{RE}_n} + 1} \right\} \right), \tag{14}$$

where the superscript co denotes the colluding.

In this system, the RSU cooperates with the PT and vehicles so that the artificial noises generated by RSU can be removed at legitimate receivers [32], [33]. The achievable secrecy rates of confidential data during the broadcast charging request with non-colluding and colluding cases can be expressed, respectively, as

$$R_1^{nc} = R_{\mathrm{VT}} - R_{\mathrm{VE}}^{nc}, \tag{15}$$

$$R_1^{co} = R_{\mathrm{VT}} - R_{\mathrm{VE}}^{co}. \tag{16}$$

The PT receives the requested message from the EV, then decodes the received message and processes it with MAC verification and decryption. Once successfully checked, the PT transmits the acknowledgment back to the EV to start the dynamic charging. The achievable rate of the $\mathrm{PT} - \mathrm{EV}$ link is expressed as

$$R_{\mathrm{TV}} = \log_2(1 + \Psi_{\mathrm{T}} \gamma_{\mathrm{TV}}). \tag{17}$$

Under jamming signals, the achievable rate of the $\mathrm{PT} - \mathrm{Eve}$ link for non-colluding and colluding cases can be expressed, respectively, as

$$R_{\mathrm{TE}}^{nc} = \log_2 \left(1 + \max_{n=1,\ldots,N} \left\{ \frac{\Psi_{\mathrm{T}} \gamma_{\mathrm{TE}_n}}{\sum_{k=1}^{K} \Psi_{\mathrm{R}} \gamma_{\mathrm{RE}_n} + 1} \right\} \right), \tag{18}$$

$$R_{\mathrm{TE}}^{co} = \log_2 \left(1 + \sum_{n}^{N} \left\{ \frac{\Psi_{\mathrm{T}} \gamma_{\mathrm{TE}_n}}{\sum_{k=1}^{K} \Psi_{\mathrm{R}} \gamma_{\mathrm{RE}_n} + 1} \right\} \right), \tag{19}$$

where $\Psi_{\mathrm{T}} = P_{\mathrm{T}}/\sigma^2$.

The achievable secrecy rates of confidential data during the feedback charging request for non-colluding and colluding cases can be expressed, respectively, as

$$R_2^{nc} = R_{\mathrm{TV}} - R_{\mathrm{TE}}^{nc}, \tag{20}$$

$$R_2^{co} = R_{\mathrm{TV}} - R_{\mathrm{TE}}^{co}. \tag{21}$$

The randomize-and-forward strategy is applied to enhance the achievable secrecy rate at the physical layer, where the PT and vehicle encode the message with random code-books to prevent the eavesdropper from combining the messages transmitted by PT and vehicle [34], [35]. Thus, the end-to-end secrecy rate can be calculated as

$$R_{e2e}^{\mathrm{x}} = [\min(R_1^{\mathrm{x}}, R_2^{\mathrm{x}})]^+, \tag{22}$$

where $[x]^+ = \max\{x, 0\}$ and $\mathrm{x} \in \{co, nc\}$.

Once the authentication between EV and PT is completed, the EV sends a message to inform RSU to deactivate the jamming signals.

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

B. SOP Analysis for Non-Colluding Case

In this section, the secrecy outage probability of the proposed system with the non-colluding case is analyzed and evaluated. The SOP is defined by the probability that the end-to-end secrecy rate is lower than a predefined rate, $R_{th} > 0$, and is given as [36]

$$\mathrm{SOP}_{e2e}^{nc} = \Pr(R_{e2e}^{nc} < R_{th}). \tag{23}$$

Theorem 1: The exact closed-form expression for the SOP of the considered scenario with non-colluding eavesdroppers can be expressed as

$$\mathrm{SOP}_{e2e}^{nc} = 1 - (1 - I_1)(1 - I_2), \tag{24}$$

where I_1 and I_2 can be expressed as

$$\begin{aligned} I_1 = 1 - &\exp\left(-\frac{\gamma_{th}-1}{\lambda_{\mathrm{VT}}\Psi_{\mathrm{V}}}\right)\left[1 - \frac{\gamma_{th}}{\lambda_{\mathrm{VT}}\Psi_{\mathrm{V}}}\sum_{n=1}^{N}\binom{N}{n}(-1)^{n-1}\right. \\ &\times\left(\frac{\Psi_{\mathrm{V}}\lambda_{\mathrm{VE}}}{n\Psi_{\mathrm{R}}\lambda_{\mathrm{R}}}\right)^{K}\frac{(-\Delta_1)^{K-1}}{(K-1)!}\left[\sum_{k=1}^{K}\left(-\frac{\Psi_{\mathrm{V}}\Delta_1\lambda_{\mathrm{VE}}}{n\Psi_{\mathrm{R}}\lambda_{\mathrm{R}}}\right)^{-k}\right. \\ &\left.\left.\times(k-1)! - \exp\left(\frac{\Psi_{\mathrm{V}}\Delta_1\lambda_{\mathrm{VE}}}{n\Psi_{\mathrm{R}}\lambda_{\mathrm{R}}}\right)\mathrm{Ei}\left(-\frac{\Psi_{\mathrm{V}}\Delta_1\lambda_{\mathrm{VE}}}{n\Psi_{\mathrm{R}}\lambda_{\mathrm{R}}}\right)\right]\right], \end{aligned} \tag{25}$$

$$\begin{aligned} I_2 = 1 - &\exp\left(-\frac{\gamma_{th}-1}{\lambda_{\mathrm{TV}}\Psi_{\mathrm{T}}}\right)\left[1 - \frac{\gamma_{th}}{\lambda_{\mathrm{TV}}\Psi_{\mathrm{T}}}\sum_{i=1}^{N}\binom{N}{i}(-1)^{i-1}\right. \\ &\times\left(\frac{\Psi_{\mathrm{T}}\lambda_{\mathrm{TE}}}{i\Psi_{\mathrm{R}}\lambda_{\mathrm{R}}}\right)^{K}\frac{(-\Delta_2)^{K-1}}{(K-1)!}\left[\sum_{k=1}^{K}\left(-\frac{\Delta_2\Psi_{\mathrm{T}}\lambda_{\mathrm{TE}}}{i\Psi_{\mathrm{R}}\lambda_{\mathrm{R}}}\right)^{-k}\right. \\ &\left.\left.\times(k-1)! - \exp\left(\frac{\Delta_2\lambda_{\mathrm{TE}}}{i\lambda_{\mathrm{R}}}\right)\mathrm{Ei}\left(-\frac{\Delta_2\lambda_{\mathrm{TE}}}{i\lambda_{\mathrm{R}}}\right)\right]\right], \end{aligned} \tag{26}$$

wherein $\Delta_1 = \frac{\gamma_{th}}{\Psi_{\mathrm{V}}\lambda_{\mathrm{VC}}} + \frac{n}{\Psi_{\mathrm{V}}\lambda_{\mathrm{VE}}}$, $\Delta_2 = \frac{\gamma_{th}}{\Psi_{\mathrm{T}}\lambda_{\mathrm{TV}}} + \frac{i}{\Psi_{\mathrm{T}}\lambda_{\mathrm{TE}}}$, and $\gamma_{th} = 2^{R_{th}}$.

Proof: The proof of Theorem 1 is in Appendix A. ■

C. Intercept Probability Analysis in Non-Colluding Case

The intercept probability (IP) is one of the key performance metrics to indicate the secrecy performance of wireless communication systems. The IP is defined as the probability that the eavesdropper succeeds in intercepting the signals intended for the legitimate receivers [37], [38]. The IP at the Eve can be mathematically expressed as

$$\mathrm{IP}^{nc} = \Pr(\max\{R_{\mathrm{VE}}^{nc}, R_{\mathrm{TE}}^{nc}\} > R_{th}). \tag{27}$$

Theorem 2: The exact closed-form expression for the IP of the considered scenario over Rayleigh fading channels can be expressed as

$$\begin{aligned} \mathrm{IP}^{nc} = &\sum_{n=1}^{N}\binom{N}{n}(-1)^{n-1}\exp\left(-\frac{n\gamma_{th}}{\Psi_{\mathrm{V}}\lambda_{\mathrm{VE}}}\right)\left(\frac{n\Psi_{\mathrm{R}}\lambda_{\mathrm{R}}\gamma_{th}}{\Psi_{\mathrm{V}}\lambda_{\mathrm{VE}}}+1\right)^{-K} \\ &+\sum_{i=1}^{N}\binom{N}{i}(-1)^{i-1}\exp\left(-\frac{i\gamma_{th}}{\Psi_{\mathrm{T}}\lambda_{\mathrm{TE}}}\right)\left(\frac{i\Psi_{\mathrm{R}}\lambda_{\mathrm{R}}\gamma_{th}}{\Psi_{\mathrm{T}}\lambda_{\mathrm{TE}}}+1\right)^{-K} \\ &-\sum_{n=1}^{N}\sum_{i=1}^{N}\binom{N}{n}\binom{N}{i}\exp\left(-\frac{n\gamma_{th}}{\Psi_{\mathrm{V}}\lambda_{\mathrm{VE}}}-\frac{i\gamma_{th}}{\Psi_{\mathrm{T}}\lambda_{\mathrm{TE}}}\right) \\ &\times(-1)^{i+n-2}\left(1+\frac{n\Psi_{\mathrm{R}}\lambda_{\mathrm{R}}\gamma_{th}}{\Psi_{\mathrm{V}}\lambda_{\mathrm{VE}}}+\frac{i\Psi_{\mathrm{R}}\lambda_{\mathrm{R}}\gamma_{th}}{\Psi_{\mathrm{T}}\lambda_{\mathrm{TE}}}\right)^{-K}, \end{aligned} \tag{28}$$

where $\gamma_{th} \triangleq 2^{R_{th}} - 1$.

Proof: The proof of Theorem 2 is in Appendix B. ■

D. SOP and IP Expressions in Colluding Case

The SOP and IP at the Eve for colluding case can be expressed, respectively, as

$$\mathrm{SOP}_{e2e}^{co} = \Pr(R_{e2e}^{co} < R_{th}), \tag{29}$$

$$\mathrm{IP}^{co} = \Pr(\max\{R_{\mathrm{VE}}^{co}, R_{\mathrm{TE}}^{co}\} > R_{th}). \tag{30}$$

In the colluding case, the SOP and IP analyses have no closed-form expression based on our best experiments. Thus, the colluding case is demonstrated by Monte-Carlo simulations, which will be presented in the next section.

IV. Security Analysis

A. Impersonation Attacks

The designed DoES-SEM protocol suite can protect the EV from impersonation attacks.

Proof: To impersonate a legitimate EV to obtain the session keys, Eve broadcasts its public keys to its surrounding EVs with a strong signal power to make these EVs believe the public key belongs to the nearest CS. To collect information about EVs, Eve now plays the malicious CS and keeps sending the information update request messages to any surrounding EVs. By employing the proposed DoES method, the information update message is first encrypted by Eve's public key, then signed by EV's private key, and finally outer encrypted by Eve's public key. If Eve catches the message and performs outer decryption for the received message, it has to remove the signature of EV before inner decrypting the message. After obtaining the identity and battery status of the EV, Eve constructs the authentication request, which is first encrypted by CS's public key, signed by Eve's private key, and outer encrypted by another CS's public key following the format and encryption as shown in Fig. 4 of the revised manuscript. Once receiving the message, the CS decrypts it and fails to verify the signature since the signature did not come from the EV. Thus, the CS ignores the received message even if the identity of the EV is valid. This proof demonstrates that the proposed DoES protocol can resist EV impersonation attacks.

B. Man-in-The-Middle Attacks

The designed DoES-SEM protocol suite can protect the EV from man-in-the-middle attacks.

Proof: When data is transmitted over a shared communication channel and Eve can capture the exchange messages between EV and CS to obtain the IDs of EVs and gain benefits or to accumulate EV's IDs preparing for the DoS attacks. However, by using a one-way hash function during the signing process

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

along with additional outer encryption. Eve cannot find any vital information since it does know the keys and their sizes to decipher the communication between the EV and the CS as shown in Fig. 4(b), and these parameters are computed on a session-by-session basis. In the worst case, the Eve can perform the outer decryption and has to remove the signature of EV before decrypting the message to obtain the EV's ID. When the signature of EV is removed from the message, the integrity is not guaranteed, thus causing the failure of the authentication process. As a result, it can be highlighted that the proposed protocol is resistant to man-in-the-middle attacks.

C. Replay Attacks

The designed DoES-SEM protocol suite can protect the EV from replay attacks.

Proof: Eve may capture the message AutReq, as shown in (1), sent from EV to the CS. He could replay the AutReq message to the CS and impersonate the EV. However, the DoES scheme does not allow Eve to perform the inner decryption without removing the signature of EV, thus protecting the integrity of the EV-CS communication. Moreover, when Eve captures the CharReq message, as shown in (6), Eve needs to remove Tag and calculate the receiver tag of ciphertext by using session keys within the accepted time mismatch. Otherwise, the authentication will be corrupted. This way, the DoES scheme prevents replay attacks.

D. DoS Attacks

The designed DoES protocol can protect the EV from DoS attacks.

Proof: Eve can perform the DoS attacks to flood the system and reduce the CS's performance by sending multiple request messages with varying identities collected from the information update messages of EVs. The CS will be busy verifying those identities, which leads to resource unavailability at the CS and causes a delay in the verification and failure of the authentication process. The proposed DoES protocol is used for EV-CS authentication, where the sending and receiving messages are double encrypted and signed. Thus, CS thoroughly checks the legitimacy of received messages and discards them immediately if they are illegitimate. Therefore, the proposed DoES protocol prevents Eve from collecting the legal identity of EVs, which is the initial step of DoS attacks.

E. Forward Security

The DoES-SEM protocol suite achieves forward secrecy.

Proof: The proposed protocol ensures that session keys cannot be computed in the future, even if the long-term session key credentials are revealed. By using two encryption layers for sending public keys as shown in (1) of the DoES scheme, the secret session keys are guaranteed at the receiver and it is hard to obtain/compute these keys. Moreover, the session keys are selected from a set of random numbers in each session along with the timestamp. In the SEM scheme, session keys will be removed from each device's memory at the end of each authentication session, thus preventing attackers from using the previous sessions to compromise communications in the next sessions. Therefore, the DoES-SEM protocol suite satisfies the requirements for forward secrecy.

F. Eavesdropping At the Physical Layer

We propose an artificial noise-based security scheme to protect EV-PT communication from eavesdropping attacks at the physical layer. Eve will intercept the information exchanged between EV and CS. The RSU will generate artificial noise to reduce the quality of signals received at the Eve, thus reducing the probability of wiretapping channels. We have shown in the numerical results that the proposed AN-based security scheme provides the highest performance among other schemes in dealing with eavesdropper attacks.

Table I shows the comparison of security features between the proposed protocol and its relevant counterparts in [16], [17], [18], [19], [20], [21]. Besides securing the authentication process, the proposed protocol with PHY security feature also provides security at the physical layer by using artificial noise to reduce the wiretapped signals at the eavesdroppers and increase the level of security for EV dynamic charging systems.

V. Numerical Results

In this section, we present illustrative numerical results for the achievable performance of the proposed authentication and AN-based PLS schemes. We implement the proposed DoES and SEM schemes on the C++ platform using Crypto++ 8.6 library while the AN-based PLS scheme is implemented in Matlab using Monte-Carlo simulations with 10^6 independent trials. In the proposed DoES scheme, the ECDSA, which has been recommended by the IEEE 802.11p standard for authenticating vehicular communications [39], is used for signing while the Rivest–Shamir–Adleman (RSA) algorithm is used for asymmetric encryption. In the SEM scheme, the ECDSA is used for signing, the Advanced Encryption Standard (AES) is used for symmetric encryption, and the Cipher-based MAC (CMAC) is used for verifying both the integrity and authenticity of messages. Unless otherwise stated, we set the key sizes for ECDSA, RSA, AES, and CMAC as 256, 1024, 128, and 128 bits, respectively. For the AN-based PLS setting, when considering one coherent time slot, the distances among EV, PT, and eavesdroppers are fixed. Thus, the distances of the EV-PT, EV-Eve, and PT-Eve are 10 m, 24 m, and 24 m, respectively. The reference pathloss is $\sigma^{PL} = -30$ dB, the number of eavesdroppers $N = 5$, the number of antennas at RSU $K = 3$, the pathloss exponent $\beta = 3$, the normalized noise variance $\sigma^2 = 1$, the rate threshold $R_{th} = 1$.

We first examine the computation overhead indicating the execution time of cryptographic functions used for DoES and SEM schemes to process the messages, as shown in Table II . As observed, asymmetric cryptography such as RSA, and ECDSA, is slower and takes a longer time to execute the message than symmetric ones such as AES and CMAC. This is due to the fact that the former uses longer keys than the latter one which contributes to slower encryption speed. Therefore, the RSA and ECDSA are used for designing the DoES scheme to exchange

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SECURITY FEATURES COMPARISON WITH RELEVANT PROTOCOLS

Authentication Protocols	Impersonation	Replay	Man-in-the-middle	DoS	Foward security	PHY security
Proposed Protocol	✓	✓	✓	✓	✓	✓
Protocol in [16]	✓	✓	✓	✓	✓	✗
Protocol in [17]	✓	✓	✓	✓	✓	✗
Protocol in [18]	✓	✓	✓	✗	✗	✗
Protocol in [19]	✓	✓	✓	✗	✓	✗
Protocol in [20]	✓	✓	✓	✗	✓	✗
Protocol in [21]	✓	✓	✓	✗	✗	✗

TABLE II
AVERAGE EXECUTION TIME OF EACH FUNCTION

Function	Time (ms)
RSA AEn	4
RSA Decryption	0.24
ECDSA Sign	3
ECDSA Verification	0.475
AES En	7.55×10^{-3}
AES Decryption	3.733×10^{-3}
CMAC	2.45×10^{-3}
CMAC Verification	3.79×10^{-3}

TABLE III
AVERAGE EXECUTION TIME (IN MILLISECONDS) OF DOES SCHEME WITH DIFFERENT RSA KEY SIZES

RSA key (bit) for DoES Scheme	Execution time at the sender	Execution time at the receiver
1024	25 ms	2 ms
2048	195 ms	8 ms
4096	2038 ms	46 ms

TABLE IV
AVERAGE EXECUTION TIME (IN MILLISECOND) OF SEM SCHEME WITH DIFFERENT AES KEY SIZES

AES key (bit) for SEM Scheme	Execution time at the transmitter	Execution time at the receiver
128	3.51 ms	0.51 ms
192	3.85 ms	0.56 ms
256	3.91 ms	0.57 ms

TABLE V
AVERAGE EXECUTION TIME OF THE PROPOSED SCHEMES AND THE SCHEME IN [6]

Authentication	Proposed DoES and SEM Schemes	The Scheme in [6]
EV-CS	52 ms	50 ms
EV-PT (or CPs)	8.23 ms	37 ms
Total	60.23 ms	87 ms

session keys before the charging process while the AES and CMAC are used for designing the SEM scheme to perform online authentication.

In Table III, we show the effect of key sizes on the execution time of the proposed DoES scheme operating at the sender and receiver sides. As can be seen the bigger the size of the encryption key, the longer the execution time that the DoES scheme takes. In addition, the DoES scheme on the sender side requires generating two pairs of private and public keys for double encryption and a pair of keys for signature generation, thus the encryption process at the sender is generally slower than decryption at the receiver side.

The effect of key sizes on the execution time of the proposed SEM Scheme is shown in Table IV. As observed, the SEM scheme with 128 is slightly faster than that of 192 and 256 bits on the sender and receiver sides. However, the AES with a 256-bit key can resist the brute force key guessing attacks more than its counterpart with a 128-bit secret key. The sender side takes much more time for processing messages than the receiver side since the SEM scheme spends time generating the secret session keys and the initialization vector for the AES algorithm and CMAC.

Table V shows that the proposed DoES and SEM schemes just spend 52 ms for EV-CS authentication and 8.23 ms for EV-PT authentication, respectively, which can meet the delay requirement for online authentications between EV and PT in DWPT systems. On the other hand, the authentication scheme in [6] is slower than the proposed SEM scheme with an execution time of 37 ms for EV-CPs authentication, which may be difficult for online authentication when EV approaches the charging lanes.

As can be observed in Table VI, the proposed protocol requires a higher level of communication overheads (2816 bytes) than other existing ones. However, the proposed DoES-SEM protocol suite provides a comprehensive high-security level compared to other existing ones, especially the PHY security feature to protect the EV charging process at the physical layer. This shows the benefits of deploying the DoES-SEM protocol suit in dealing with various attacks at the level of network and physical layers in EV dynamic charging systems.

Fig. 7 shows the SOP of the proposed EV system versus Ψ_{V} with different benchmark security schemes. As observed, when the average received SNRs at the EV and PT increase, the SOP

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMMUNICATION COSTS COMPARISON WITH OTHER PROTOCOLS IN EACH SESSION

Authentication Protocols	Number of messages	Communication overhead (bytes)
Proposed Protocol (DoES+SEM)	9	2816
Authentication protocol in [16]	7	164
Authentication protocol in [6]	9	640
Authentication protocol in [40]	5	698

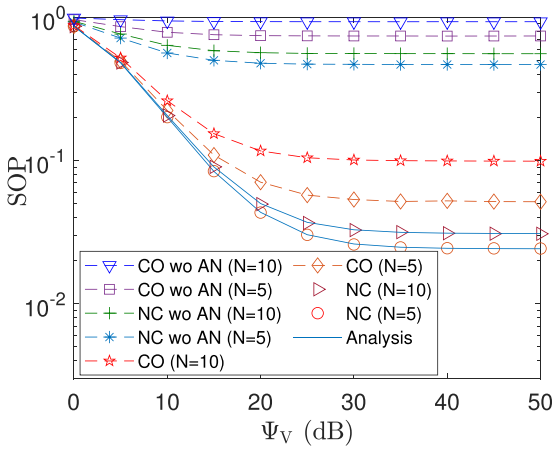


Fig. 7. Proposed AN-based PLS scheme and its counterpart without artificial noise.

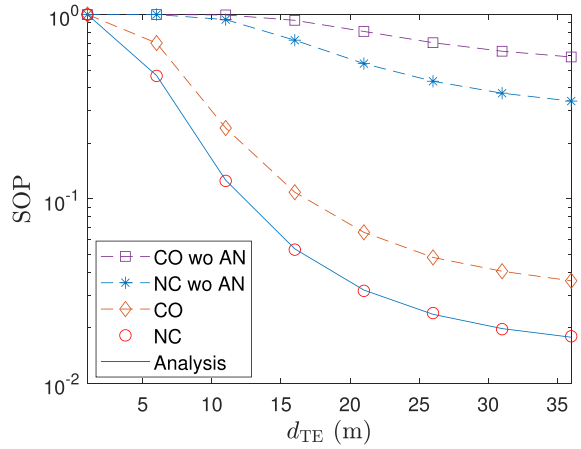


Fig. 8. Proposed AN-based PLS scheme and its counterpart without artificial noise.

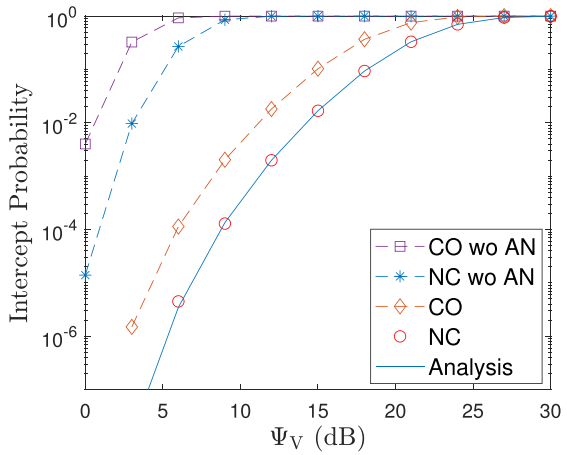


Fig. 9. Proposed AN-based PLS scheme and its counterpart without artificial noise.

performance of all schemes is improved and eventually saturates at its given secrecy outage floors at high SNRs. The reason is that the increase in the average SNR of the main channel can lower the SOP, thus making the system more secure. However, further increasing Ψ_V will not bring any additional benefits to the security performance because the average wiretapping SNR will also increase. This reveals that the eavesdropper does not generally require very high average transmit SNRs to successfully eavesdrop on ongoing transmissions. The system with non-colluding has better SOP performance than its counterpart with colluding. It is because all eavesdroppers collaborate in colluding mode to achieve the best version of wiretapped messages while only the strongest Eve overhears the legitimate messages in the non-colluding case. When the number of eavesdroppers increases, the SOP gap of colluding mode is larger than that of the non-colluding one which shows the colluding case is more prone to the eavesdropping attack than the non-colluding one. Moreover, the SOPs of all scenarios increase with N since deploying more eavesdroppers can enhance the performance of illegitimate eavesdroppers and increase the possibility of wiretapping messages exchanged between the EV and PT. Furthermore, the proposed AN-based scheme with NC and CO cases yields higher SOP performance than its counterpart without AN, which shows the important role of RSU in generating AN to prevent DWPT systems from eavesdropping attacks.

In Fig. 8, we show the effect of the distance between Eve and the power transmitter on the SOP of all scenarios. It is clear that when the eavesdroppers move further away from the PT, the SOP of all scenarios is improved. The reason is that when d_{TE} increases, the wiretap channel gains worsen due to the path loss effect, which also deteriorates the received SNR at Eves. Again, the AN-based PLS scenarios show the advantage of using AN for degrading the signal quality at eavesdroppers, which makes the SOP of the system with AN smaller than that of the system without using AN. Moreover, the analysis results of the SOP in the non-colluding case perfectly match the simulation ones, which demonstrates the correctness of the developed analysis in Theorem 1.

Fig. 9 shows the system IP versus Ψ_V with different scenarios. As observed, the IP of all scenarios increases as Ψ_V grows since increasing the received SNRs at the EV means the probability that eavesdroppers can capture exchanged messages for data recovery is high. The CO wo AN scenario has the highest probability of being intercepted, which is the most vulnerable to

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

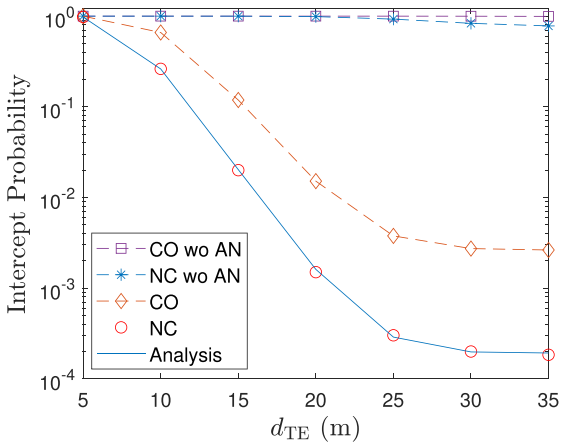


Fig. 10. Proposed AN-based PLS scheme and its counterpart without artificial noise.

the eavesdropping attack. On the other hand, the non-colluding scenario with AN can achieve the lowest intercept probability, explicitly demonstrating the security and reliability benefits of exploiting artificial noise for degrading the channel quality of eavesdroppers.

In Fig. 10, we show the effect of distance between Eve and the PT on the IP of all scenarios. As observed, the system (colluding and non-colluding) without AN cannot withstand the eavesdropper attack regardless of the distance between Eve and PT. Again, the proposed system with AN has the lowest intercept probability, especially when Eve is located further away from the PT. Finally, the analysis results for the non-colluding case in Figs. 7, 8, 9, and 10 are in excellent agreement with the simulation ones, validating the correctness of our derivations in Theorems 1 and 2.

VI. Conclusion

In this article, we proposed the DoES and SEM authentications and AN-based PLS schemes for strengthening dynamic wireless power transfer systems. The DoES scheme was proven to resist various attacks in DWPT systems such as man-in-the-middle attacks, replay attacks, forward secrecy attacks, insider attacks, and eavesdropping threats while the SEM was shown to guarantee low latency, privacy requirements, and resistance to tampering attacks. The proposed AN-based PLS scheme could decrease the signal quality received at non-colluding eavesdroppers with low SOP and IP, which protected the DWPT system from eavesdropping attacks. Closed-form expressions for the SOP and IP of the system with the non-colluding case were derived to show some insights into the physical security characteristics of DWPT systems. Numerical results show the effectiveness of the proposed DoES and SEM schemes with low latency authentication, which allows exploiting the potential of such schemes for DWPT systems. In our future works, we will consider blockchain technology to enhance security and provide a decentralized and transparent method for verifying the identity of EVs. By checking the blockchain records, the EVs are only granted access to the communication network with RSU, PT, and CS if the EV's identity is valid and not compromised.

Appendix A
Proof of Theorem 1

From (23), the end-to-end SOP can be rewritten as follows:

$$\begin{aligned}\mathrm{SOP}_{e2e} &= \Pr\left[\min\left\{\frac{1+\Psi_{\mathrm{V}}\gamma_{\mathrm{VC}}}{1+X_1}, \frac{1+\Psi_{\mathrm{C}}\gamma_{\mathrm{CV}}}{1+X_2}\right\} < \gamma_{th}\right] \\ &= 1 - \Bigg(1 - \underbrace{\Pr\left[\frac{1+\Psi_{\mathrm{V}}\gamma_{\mathrm{VC}}}{1+X_1} < \gamma_{th}\right]}_{I_1}\Bigg) \\ &\quad \times \Bigg(1 - \underbrace{\Pr\left[\frac{1+\Psi_{\mathrm{C}}\gamma_{\mathrm{CV}}}{1+X_2}\right]}_{I_2}\Bigg) \end{aligned} \tag{31}$$

where $X_1 \triangleq \max\limits_{n=1,\ldots,N}\left\{\frac{\Psi_{\mathrm{V}}\gamma_{\mathrm{VE}_n}}{\sum_{k=1}^{K}\Psi_{\mathrm{R}}\gamma_{\mathrm{R}_k}+1}\right\}$ and $X_2 \triangleq \max\limits_{i=1,\ldots,N}\left\{\frac{\Psi_{\mathrm{C}}\gamma_{\mathrm{CE}_i}}{\sum_{k=1}^{K}\Psi_{\mathrm{R}}\gamma_{\mathrm{R}_k}+1}\right\}$.

The probability in I_1 can be calculated as

$$\begin{aligned} I_1 &= \int_0^{+\infty} F_{\gamma_{\mathrm{VC}}}\left(\frac{\gamma_{th}-1}{\Psi_{\mathrm{V}}} + \frac{\gamma_{th}y}{\Psi_{\mathrm{V}}}\right) f_{X_1}(y)dy \\ &= \int_0^{+\infty}\left[1 - \exp\left(-\frac{\gamma_{th}-1}{\Psi_{\mathrm{V}}\lambda_{\mathrm{VC}}}\right)\exp\left(-\frac{\gamma_{th}y}{\Psi_{\mathrm{V}}\lambda_{\mathrm{VC}}}\right)\right] f_{X_1}(y)dy. \end{aligned} \tag{32}$$

Considering X_1, let $Y \triangleq \sum_{k=1}^{K}\gamma_{\mathrm{R}_k}$ and $Z_{\mathrm{VE}} \triangleq \max\limits_{j=1,\ldots,J}\gamma_{\mathrm{VE}_j}$. Since Y is the summation of K independent exponential RVs, it follows the chi-square distribution. The PDF of Y is given as [41, Eq. (26)]

$$f_Y(x) = \left(\frac{1}{\lambda_{\mathrm{R}}}\right)^K \frac{x^{K-1}}{\Gamma(K)}\exp\left(-\frac{x}{\lambda_{\mathrm{R}}}\right). \tag{33}$$

Since Z_{VE} presents the maximum of J independent exponential random variables (RVs), the CDF of Z_{VE} is given as [41, Eq. (28)]

$$F_{Z_{\mathrm{VE}}}(x) = 1 - \sum_{n=1}^{N}\binom{N}{n}(-1)^{n-1}\exp\left(-\frac{nx}{\lambda_{\mathrm{VE}}}\right). \tag{34}$$

The CDF of X_1 can be calculated as

$$F_{X_1}(x) = \int_0^{+\infty} F_Y\left(\frac{x}{\Psi_{\mathrm{V}}} + xy\right) f_Z(y)dy \tag{35}$$

Plugging $F_{Z_{\mathrm{VE}}}(.)$ and $f_Y(.)$ into (35) and after some mathematical manipulations, we obtain as

$$\begin{aligned} F_{X_1}(x) &= 1 - \sum_{n=1}^{N}\binom{N}{n}(-1)^{n-1}\exp\left(-\frac{nx}{\Psi_{\mathrm{V}}\lambda_{\mathrm{VE}}}\right) \\ &\quad \times \left(\frac{\Psi_{\mathrm{V}}\lambda_{\mathrm{VE}}}{\Psi_{\mathrm{R}}n\lambda_{\mathrm{R}}}\right)^K \left(x + \frac{\Psi_{\mathrm{V}}\lambda_{\mathrm{VE}}}{\Psi_{\mathrm{R}}n\lambda_{\mathrm{R}}}\right)^{-K}. \end{aligned} \tag{36}$$

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

Next, applying partial integral method for I_1 in (32), then substituting (36) into the obtained equation, we have

$$I_1 = 1 - \exp\left(-\frac{\gamma_{th}-1}{\Psi_{\rm V}\lambda_{\rm VC}}\right)\left[1 - \sum_{n=1}^{N}\binom{N}{n}(-1)^{n-1}\frac{\gamma_{th}}{\Psi_{\rm V}\lambda_{VC}} \times \left(\frac{\Psi_{\rm V}\lambda_{\rm VE}}{\Psi_{\rm R}n\lambda_{RE}}\right)^{K}\int_0^{+\infty}\exp(-\Delta_1 y)\left(y+\frac{\Psi_{\rm V}\lambda_{\rm VE}}{\Psi_{\rm R}n\lambda_{\rm RE}}\right)^{-K}dy\right], \quad (37)$$

where Δ_1 has been defined in (26). Making use of [42, Eq. (3.353.2)] for the integral in (37), we obtain the desired results as (25). Similar to the derived steps for I_1, I_2 can be obtained as (26), which completes the proof of Theorem 1.

Appendix B
Proof of Theorem 2

From (27), the IP can be rewritten as

$$\begin{aligned}{\rm IP} &= 1 - \Pr\left[\max\left\{\max_{n=1,\ldots,N}\left\{\frac{\Psi_{\rm V}\gamma_{{\rm VE}_n}}{\sum_{k=1}^{K}\Psi_{\rm R}\gamma_{{\rm R}_k}+1}\right\}, \max_{i=1,\ldots,N}\left\{\frac{\Psi_{\rm T}\gamma_{{\rm TE}_i}}{\sum_{k=1}^{K}\Psi_{\rm R}\gamma_{{\rm R}_k}+1}\right\}\right\} < \gamma_{th}\right], \\ &= 1 - \Pr\left[\frac{\Psi_{\rm V}Z_{\rm VE}}{1+\Psi_{\rm R}Y} < \gamma_{th}, \frac{\Psi_{\rm T}Z_{\rm TE}}{1+\Psi_{\rm R}Y} < \gamma_{th}\right], \end{aligned} \quad (38)$$

where $Z_{\rm TE} \triangleq \max_{i=1,\ldots,N}\gamma_{{\rm TE}_i}$ and it has the CDF as [41, Eq. (28)]

$$F_{Z_{\rm TE}}(x) = 1 - \sum_{i=1}^{N}\binom{N}{i}(-1)^{i-1}\exp\left(-\frac{ix}{\lambda_{\rm TE}}\right). \quad (39)$$

From (38), the IP can be further calculated as

$$\begin{aligned}{\rm IP} &= 1 - \Pr\left[Z_{\rm VE} < \frac{\gamma_{th}}{\Psi_{\rm V}} + \frac{\gamma_{th}\Psi_{\rm R}Y}{\Psi_{\rm V}}, Z_{\rm TE} < \frac{\gamma_{th}}{\Psi_{\rm T}} + \frac{\gamma_{th}\Psi_{\rm R}Y}{\Psi_{\rm T}}\right], \\ &= 1 - \int_0^{+\infty} F_{Z_{\rm VE}}\left(\frac{\gamma_{th}}{\Psi_{\rm V}} + \frac{\gamma_{th}\Psi_{\rm R}y}{\Psi_{\rm V}}\right) \times F_{Z_{\rm TE}}\left(\frac{\gamma_{th}}{\Psi_{\rm T}} + \frac{\gamma_{th}\Psi_{\rm R}y}{\Psi_{\rm T}}\right) f_Y(y)dy. \end{aligned} \quad (40)$$

Plugging the CDF of $Z_{\rm VE}$, $Z_{\rm TE}$ and Y into (40) and after some manipulations, we obtain the desired result as (28). The proof of Theorem 2 is concluded.

References

[1] G. A. Covic and J. T. Boys, "Modern trends in inductive power transfer for transportation applications," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 1, no. 1, pp. 28–41, Mar. 2013.

[2] H. Wang, U. Pratik, A. Jovicic, N. Hasan, and Z. Pantic, "Dynamic wireless charging of medium power and speed electric vehicles," *IEEE Trans. Veh. Technol.*, vol. 70, no. 12, pp. 12552–12566, Dec. 2021.

[3] N. Chen, M. Wang, N. Zhang, and X. Shen, "Energy and information management of electric vehicular network: A survey," *IEEE Commun. Surv. Tut.*, vol. 22, no. 2, pp. 967–997, Secondquarter 2020.

[4] P. R. Babu, B. Palaniswamy, A. G. Reddy, V. Odelu, and H. S. Kim, "A survey on security challenges and protocols of electric vehicle dynamic charging system," *Secur. Privacy*, vol. 5, no. 3, 2022, Art. no. e210.

[5] H.-C. Hsiao et al., "Flooding-resilient broadcast authentication for VANETs," in *Proc. 17th Annu. Int. Conf. Mobile Comput. Netw.*, 2011, pp. 193–204.

[6] H. Li, G. Dán, and K. Nahrstedt, "Portunes+: Privacy-preserving fast authentication for dynamic electric vehicle charging," *IEEE Trans. Smart Grid*, vol. 8, no. 5, pp. 2305–2313, Sep. 2017.

[7] M. Pazos-Revilla, A. Alsharif, S. Gunukula, T. N. Guo, M. Mahmoud, and X. Shen, "Secure and privacy-preserving physical-layer-assisted scheme for EV dynamic charging system," *IEEE Trans. Veh. Technol.*, vol. 67, no. 4, pp. 3304–3318, Apr. 2018.

[8] P. Razmjouei, A. Kavousi-Fard, M. Dabbaghjamanesh, T. Jin, and W. Su, "Dag based smart contract for dynamic 6G wireless EVs charging system," *IEEE Trans. Green Commun. Netw.*, vol. 6, no. 3, pp. 1459–1467, Sep. 2022.

[9] M. Abouyoussef and M. Ismail, "Blockchain-based privacy-preserving networking strategy for dynamic wireless charging of EVs," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 2, pp. 1203–1215, Jun. 2022.

[10] G. Kumar et al., "A privacy-preserving secure framework for electric vehicles in IoT using matching market and signcryption," *IEEE Trans. Veh. Technol.*, vol. 69, no. 7, pp. 7707–7722, Jul. 2020.

[11] H. Li, G. Dán, and K. Nahrstedt, "Proactive key dissemination-based fast authentication for in-motion inductive EV charging," in *Proc. IEEE Int. Conf. Commun.*, 2015, pp. 795–801.

[12] R. Hussain, D. Kim, M. Nogueira, J. Son, A. Tokuta, and H. Oh, "A new privacy-aware mutual authentication mechanism for charging-on-the-move in online electric vehicles," in *Proc. 11th Int. Conf. Mobile Ad-hoc Sensor Netw.*, 2015, pp. 108–115.

[13] S. Gunukula, A. B. Sherif, M. Pazos-Revilla, B. Ausby, M. Mahmoud, and X. S. Shen, "Efficient scheme for secure and privacy-preserving electric vehicle dynamic charging system," in *Proc. IEEE Int. Conf. Commun.*, 2017, pp. 1–6.

[14] T. Zhao, L. Wei, and C. Zhang, "A secure and privacy-preserving billing scheme for online electric vehicles," in *Proc. IEEE 83rd Veh. Technol. Conf.*, 2016, pp. 1–5.

[15] M. Tajmohammadi, S. M. Mazinani, M. Nikooghadam, and Z. Al-Hamdawee, "LSPP: Lightweight and secure payment protocol for dynamic wireless charging of electric vehicles in vehicular cloud," *IEEE Access*, vol. 7, pp. 148424–148438, 2019.

[16] P. R. Babu, A. G. Reddy, B. Palaniswamy, and S. K. Kommuri, "EV-Auth: Lightweight authentication protocol suite for dynamic charging system of electric vehicles with seamless handover," *IEEE Trans. Intell. Veh.*, vol. 7, no. 3, pp. 734–747, Sep. 2022.

[17] P. R. Babu, R. Amin, A. G. Reddy, A. K. Das, W. Susilo, and Y. Park, "Robust authentication protocol for dynamic charging system of electric vehicles," *IEEE Trans. Veh. Technol.*, vol. 70, no. 11, pp. 11338–11351, Nov. 2021.

[18] A. Alshaeri and M. Younis, "Lightweight authentication and authorization protocol for dynamic charging of electric vehicles," in *Proc. IEEE 19th Annu. Consum. Commun. Netw. Conf.*, 2022, pp. 550–556.

[19] Y. Wang, H. T. Luan, Z. Su, N. Zhang, and A. Benslimane, "A secure and efficient wireless charging scheme for electric vehicles in vehicular energy networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 2, pp. 1491–1508, Feb. 2022.

[20] P. R. Babu, A. G. Reddy, B. Palaniswamy, and A. K. Das, "EV-PUF: Lightweight security protocol for dynamic charging system of electric vehicles using physical unclonable functions," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 5, pp. 3791–3807, Sep./Oct. 2022.

[21] E. ElGhanam, I. Ahmed, M. Hassan, and A. Osman, "Authentication and billing for dynamic wireless EV charging in an internet of electric vehicles," *Future Internet*, vol. 13, no. 10, 2021, Art. no. 257.

[22] L. Reyzin and N. Reyzin, "Better than BiBa: Short one-time signatures with fast signing and verifying," in *Proc. Australas. Conf. Inf. Secur. Privacy*, 2002, pp. 144–153.

[23] K. Hamouid and K. Adi, "Privacy-aware authentication scheme for electric vehicle in-motion wireless charging," in *Proc. Int. Symp. Netw. Comput. Commun.*, 2020, pp. 1–6.

[24] S. Garg, K. Kaur, G. Kaddoum, J. J. Rodrigues, and M. Guizani, "Secure and lightweight authentication scheme for smart metering infrastructure in smart grid," *IEEE Trans. Ind. Informat.*, vol. 16, no. 5, pp. 3548–3557, May 2020.

[25] D. Davis, "Defective sign & encrypt in S/MIME, PKCS# 7, MOSS, PEM, PGP, AND XML," in *Proc. USENIX Annu. Tech. Conf. Gen. Track*, 2001, pp. 65–78.

[26] R. Anderson and R. Needham, "Robustness principles for public key protocols," in *Proc. Annu. Int. Cryptol. Conf.*, 1995, pp. 236–247.

[27] D. Vincent, P. S. Huynh, N. A. Azeez, L. Patnaik, and S. S. Williamson, "Evolution of hybrid inductive and capacitive AC links for wireless EV charging—A comparative overview," *IEEE Trans. Transport. Electrific.*, vol. 5, no. 4, pp. 1060–1077, Dec. 2019.

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.

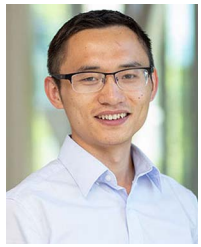
[28] A. Echols, S. Mukherjee, M. Mickelsen, and Z. Pantic, "Communication infrastructure for dynamic wireless charging of electric vehicles," in *Proc. IEEE wireless Commun. Netw. Conf.*, 2017, pp. 1–6.
[29] B. He, Y. She, and V. K. Lau, "Artificial noise injection for securing single-antenna systems," *IEEE Trans. Veh. Technol.*, vol. 66, no. 10, pp. 9577–9581, Oct. 2017.
[30] C. Shen and J. Huang, "Earfisher: Detecting wireless eavesdroppers by stimulating and sensing memory EMR," in *Proc. 18th USENIX Symp. Netw. Syst. Des. Implementation*, 2021, pp. 873–886.
[31] H. Lei et al., "Secrecy outage of max–min TAS scheme in MIMO-NOMA systems," *IEEE Trans. Veh. Technol.*, vol. 67, no. 8, pp. 6981–6990, Aug. 2018.
[32] L. Dong, Z. Han, A. P. Petropulu, and H. V. Poor, "Improving wireless physical layer security via cooperating relays," *IEEE Trans. Signal Process.*, vol. 58, no. 3, pp. 1875–1888, Mar. 2010.
[33] T. M. Hoang, T. Q. Duong, N.-S. Vo, and C. Kundu, "Physical layer security in cooperative energy harvesting networks with a friendly jammer," *IEEE Wirel. Commun. Lett.*, vol. 6, no. 2, pp. 174–177, Apr. 2017.
[34] J. Mo, M. Tao, and Y. Liu, "Relay placement for physical layer security: A secure connection perspective," *IEEE Commun. Lett.*, vol. 16, no. 6, pp. 878–881, Jun. 2012.
[35] C. T. Dung, N. T. Van, T. T. Duy, V. N. Q. Bao, and N. L. Nhat, "Security enhancement for dual-hop RF protocols with nth-best partial relay and EH-based jammer," in *Proc. Int. Conf. Commun. Manage. Telecommun.*, 2015, pp. 111–115.
[36] T.-V. Nguyen, T.-N. Tran, K. Shim, T. Huynh-The, and B. An, "Secrecy outage performance and deep learning evaluation of multihop energy harvesting IoT networks over Nakagami-m fading channels," in *Proc. Int. Symp. Comput. Consum. Control*, 2020, pp. 484–488.
[37] Y. Zou, B. Champagne, W.-P. Zhu, and L. Hanzo, "Relay-selection improves the security-reliability trade-off in cognitive radio systems," *IEEE Trans. Commun.*, vol. 63, no. 1, pp. 215–228, Jan. 2015.
[38] P. T. Tin and T. T. Duy, "Power allocation strategies for dual-hop relay protocols with best relay selection under constraint of intercept probability," *ICT Exp.*, vol. 5, no. 1, pp. 52–55, 2019.
[39] L. Chen, D. Moody, A. Regenscheid, and A. Robinson, "Digital signature standard (DSS)," *Federal Inf. Process. Stds. (NIST FIPS) - 186-5*, Nat. Inst. Standards Technol., Gaithersburg, MD, USA, 2023. [Online]. Available: https://doi.org/10.6028/NIST.FIPS.186-5
[40] K. Rabieh and M. Wei, "Efficient and privacy-aware authentication scheme for EVs pre-paid wireless charging services," in *Proc. IEEE Int. Conf. Commun.*, 2017, pp. 1–6.
[41] T.-V. Nguyen, T.-N. Do, V. N. Q. Bao, D. B. d. Costa, and B. An, "On the performance of multihop cognitive wireless powered D2D communications in WSNs," *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 2684–2699, Mar. 2020.
[42] I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series, and Products*. Cambridge, MA, USA: Academic Press, 2014.



Toan-Van Nguyen (Member, IEEE) received the B.S. degree in electronics and telecommunications engineering and the M.S. degree in electronics engineering from the HCMC University of Technology and Education, Ho Chi Minh City, Vietnam, in 2011 and 2014, respectively, and the Ph.D. degree in electronics and computer engineering from Hongik University, Sejong, South Korea, in 2021. From 2021 to 2022, he was a Postdoctoral Researcher with the Electrical and Computer Engineering Department, Utah State University, Logan, UT, USA. He is currently a Lecturer with the School of Computer Science and Engineering, International University - VNU-HCM, Thu Duc City, Ho Chi Minh City, Vietnam. His current research interests include IoT communications, beyond 5G networks, and artificial intelligence for wireless security.



Haijian Sun (Member, IEEE) received the B.S. and M.S. degrees from Xidian University, Xi'an, China, in 2011 and 2014, respectively, and the Ph.D. degree from Utah State University, Logan, UT, USA, in 2019. He is currently an Assistant Professor with the School of Electrical and Computer Engineering, The University of Georgia, Athens, GA, USA. His current research interests include wireless communication beyond 5G, machine learning at the edge, cyber security, IoT communications, wireless systems, and optimization analysis.



Hongjie Wang (Senior Member, IEEE) received the B.S. degree in electrical engineering from the University of Shanghai for Science and Technology, Shanghai, China, in 2009, the M.S. degree in electrical engineering from Shanghai Jiaotong University, Shanghai, in 2012, and the Ph.D. degree in electrical engineering from Utah State University, Logan, UT, USA, in 2018. From 2012 to 2014, he was a Senior Researcher with China Electric Power Research Institute, working on intelligent high-voltage equipment development and standardization. After graduating in 2018, he was a Postdoc Researcher with Utah State University (USU) for one year. Since 2019, he has been an Assistant Professor with Electrical and Computer Engineering Department, Utah State University. At USU, he is the Co-Director of Utah State University Power Electronics Laboratory. He has coauthored more than 50 peer-reviewed publications and six issued patents. His current research interests include dc power distribution, analysis, design, and control of resonant converters, extreme fast charging, grid interface power converter, health monitoring and diagnosis of power converters, second-life battery and battery management systems, and dynamic wireless power transfer for transportation electrification applications. Dr. Wang was the recipient of the National Science Foundation Career Award in 2023. He was also recipient of the 2022 Researcher of the Year Award and 2023 Outstanding Advisor of the Year Award from Electrical and Computer Engineering Department at Utah State University, and the 2023 Undergraduate Research Mentor of the Year Award from College of Engineering at Utah State University.



Rose Qingyang Hu (Fellow, IEEE) received the B.S. degree from the University of Science and Technology of China, Hefei, China, the M.S. degree from New York University, New York, NY, USA, and the Ph.D. degree from the University of Kansas, Lawrence, KS, USA. Besides a decade academia experience, she has more than ten years of R&D experience with Nortel, Blackberry, and Intel as a Technical Manager, a Senior Wireless System Architect, and a Senior Research Scientist, actively participating in industrial 3G/4G technology development, standardization, system level simulation, and performance evaluation. She is currently a Professor with the Electrical and Computer Engineering Department and Associate Dean for research of College of Engineering, Utah State University, Logan, UT, USA. She also directs Communications Network Innovation Lab, Utah State University. She has authored or coauthored extensively in top IEEE journals and conferences and also holds numerous patents in her research field, which include next generation wireless system design and optimization, Internet of Things, cyber physical system, mobile edge computing, V2X communications, artificial intelligence in wireless networks, wireless system modeling and performance analysis. She is currently serving on the editorial boards of IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, *IEEE Communications Magazine* and IEEE WIRELESS COMMUNICATIONS. She was also the TPC Co-Chair for the IEEE ICC 2018. She is an IEEE Communications Society Distinguished Lecturer Class during 2015–2018. She was the recipient of prestigious Best Paper Awards from IEEE GLOBECOM 2012, IEEE ICC 2015, the IEEE VTC Spring 2016, and IEEE ICC 2016. She is member of Phi Kappa Phi Honor Society.

Authorized licensed use limited to: the Leddy Library at the University of Windsor. Downloaded on October 25,2026 at 12:59:36 UTC from IEEE Xplore. Restrictions apply.